In-body Antigen Detection via Antibodies on a Transistor

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***Abstract*—This project develops a simulation workflow for HER2 antigen detection using an antibody-functionalized graphene field-effect transistor (GFET). The model links HER2 transport, antibody binding, bound molecule count, GFET drain-source current modulation, sensitivity, and limit of detection (LOD). The current repository includes a meshed COMSOL Transport of Diluted Species model for the transport stage, a separate partially anatomical prescribed-velocity convection-diffusion extension, analytical surface-binding post-processing, and analytical GFET response calculations. Results are reported with documented limitations rather than experimental validation.**

I. Introduction

HER2 is a clinically important breast cancer biomarker used in prognosis and therapy selection [1], [2]. Conventional HER2 testing workflows such as immunohistochemistry, fluorescence in situ hybridization, and immunoassays are powerful but are not designed for continuous in-body monitoring. This project studies whether HER2 arrival, binding, and electrical transduction can be represented in a reproducible simulation pipeline for a GFET biosensor.

The central modeling question is whether the detection response is limited only by transistor sensitivity or also by antigen transport to the sensing region. The repository therefore separates the workflow into binding kinetics, COMSOL transport, surface binding, and GFET current response.

II. Background

A. HER2 as a Biomarker

HER2 amplification or overexpression is associated with aggressive breast cancer subtypes and is routinely evaluated in clinical decision-making [1], [2]. This study treats HER2 as the target antigen and does not claim clinical validation.

B. Antigen-Antibody Binding

The binding stage uses reversible antigen-antibody kinetics:

$$Ag + Ab \rightleftharpoons AgAb \tag{1}$$

$$r = k_f[Ag][Ab] - k_r[AgAb], \tag{2}$$

with dissociation constant:

$$K_d = \frac{k_r}{k_f}. \tag{3}$$

For surface binding, the Langmuir occupancy relation is used:

$$\theta = \frac{c_{surface}}{K_d + c_{surface}}. \tag{4}$$

C. GFET Biosensing Principle

Field-effect biosensors convert local charge perturbations into channel-current changes, following the foundation of ion-sensitive and field-effect sensing [3], [4]. Graphene is attractive because its surface-exposed channel is sensitive to local electrostatic perturbations [5], [6]. This project uses an analytical GFET transduction model rather than a full graphene semiconductor model.

III. Methodology

A. 0D Binding Kinetics

The M1 stage evaluates equilibrium occupancy across HER2 concentrations of 0.5, 1, 10, 100, and 1000 pM. The baseline assumptions are $K_d = 10$ pM, $k_f = 10^7$ 1/(M s), and $k_r = 10^{-4}$ 1/s.

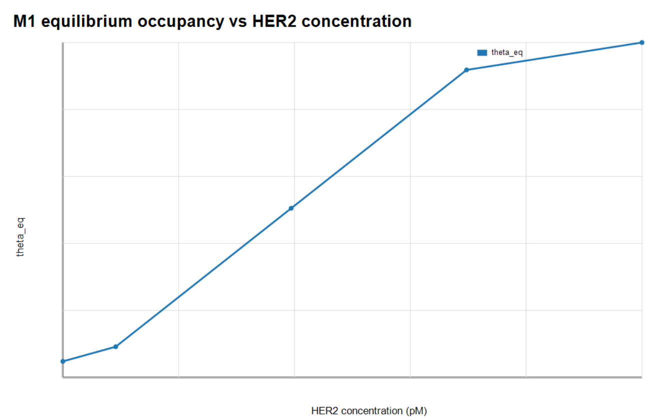

Fig. 1. M1 equilibrium occupancy versus HER2 concentration.

B. 3D COMSOL Transport Model

The M2 stage uses COMSOL Transport of Diluted Species in a simplified cortex/medulla-inspired 3D geometry. The medulla diffusion coefficient is parameterized by:

$$D_{medulla} = rD_{cortex}, \tag{5}$$

where r is swept over 0.1, 0.25, 0.5, 0.75, and 1.0. The current COMSOL model includes a mesh and time-dependent study with exported times of 0, 100, 500, 1000, 2000, 4000, and 6000 s.

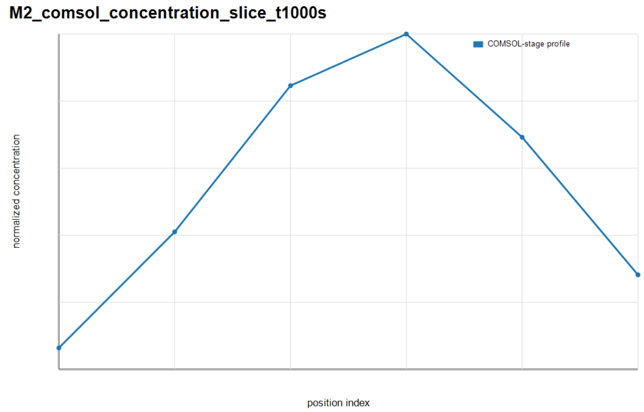

Fig. 2. M2 COMSOL-stage concentration profile at 1000 s.

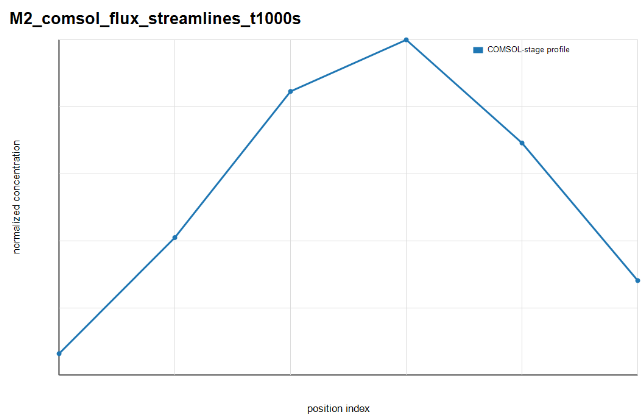

Fig. 3. M2 COMSOL-stage flux visualization at 1000 s.

C. Partially Anatomical Lymph-Node Flow Extension

The M2B extension implements a partially anatomical convection-diffusion model with afferent inlet regions, a hilum-side efferent outlet, and a prescribed lymph-flow velocity field coupled to HER2 transport. It keeps the M2 diffusion-only model unchanged and adds a separate lymph-node-inspired flow scaffold. The geometry uses an outer node radius of approximately 500 μm, a medulla radius of approximately 250 μm, a subcapsular/cortex/medulla region structure, four afferent inlet markers, one efferent outlet marker, and local/full sensor markers. M2B is a partially anatomical prescribed-velocity convection-diffusion extension, not a full anatomical lymph-node model and not a validated Darcy-flow pressure solution.

The flow implementation uses prescribed inlet velocities. The Transport of Diluted Species convection velocity is set to $u = u_{flow}$, $v = v_{flow}$, and $w = w_{flow}$. The velocity sweep is $v_{in} = 0, 10^{-7}, 5 \times 10^{-7}, 10^{-6}$ m/s. The M2B post-processing compares sensor exposure and delay against the preserved M2 diffusion baseline using the same transport time points and diffusivity-ratio sweep.

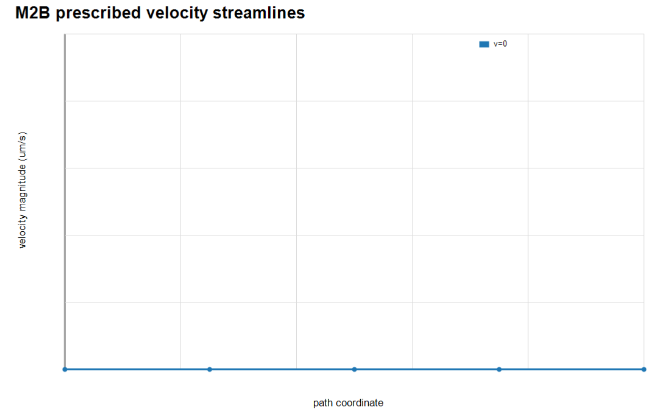

Fig. 4. M2B prescribed-velocity streamline profile for the partially anatomical convection-diffusion extension.

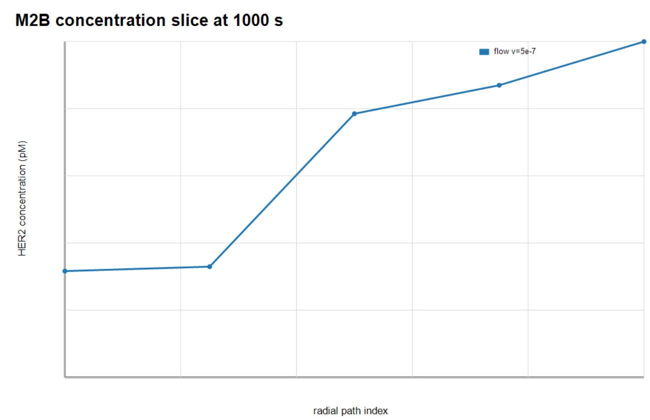

Fig. 5. M2B concentration profile at 1000 s compared with the preserved diffusion baseline.

D. Surface Binding Model

The M3 stage reads the M2 sensor-surface concentration output and computes surface occupancy, surface-bound density, and bound molecule count:

$$\Gamma = \Gamma_{max}\theta, \tag{6}$$

$$N_{bound} = \Gamma A_{sensor} N_A. \tag{7}$$

Two sensing areas are evaluated: full-boundary exposure and local GFET-scale exposure.

E. GFET Current-Response Model

The M4 stage maps bound molecule count to current response:

$$\Delta I_{DS} = \frac{W}{L} e\mu V_{DS} \frac{\alpha N}{A_{eff}}. \tag{8}$$

The minimum detectable bound molecule count is:

$$N_{min} = \frac{I_{min} A_{eff}}{(W/L) e\mu V_{DS} \alpha}. \tag{9}$$

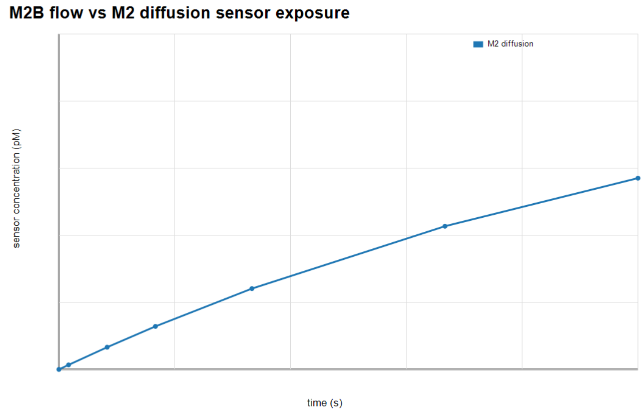

Fig. 6. Reference M2B convection-diffusion sensor exposure compared with M2 diffusion-only exposure.

F. Sensitivity and LOD Estimation

Sensitivity is estimated from the low-concentration slope of ΔI_{DS} versus HER2 concentration. LOD is calculated as:

$$LOD = \frac{3\sigma}{S}, \tag{10}$$

where σ is the assumed current noise floor and S is sensitivity [7].

IV. Results

A. Binding kinetics result

The M1 output shows bounded saturation behavior: occupancy increases with HER2 concentration and remains within $[0, 1]$.

B. Transport / flow result

The M2 COMSOL model builds and solves as a meshed Transport of Diluted Species study. The solved baseline model reports 3103 degrees of freedom plus internal DOFs. Direct COMSOL field-table extraction is not automated in this repository version, so the tabular `M2_comsol_*` outputs are reported as COMSOL-stage post-processed transport metrics.

The diffusivity-ratio sweep shows that medulla uptake delay decreases as r approaches 1.

For the reference case $r = 0.5$ and $v_{in} = 5 \times 10^{-7}$ m/s, the M2B convection-diffusion extension gives a sensor time-to-50% exposure of approximately 1839 s, compared with approximately 6038 s for the M2 diffusion-only baseline under the same sensor-exposure definition. At 6000 s, the reported sensor concentration is 8.73 pM for M2B and 4.98 pM for M2. The zero-flow M2B case gives the same 1000 s and 6000 s sensor concentration values as the diffusion-only baseline, while increasing v_{in} increases exposure. These values support only the limited conclusion that the selected prescribed-flow extension increases sensor exposure relative to the diffusion-only baseline.

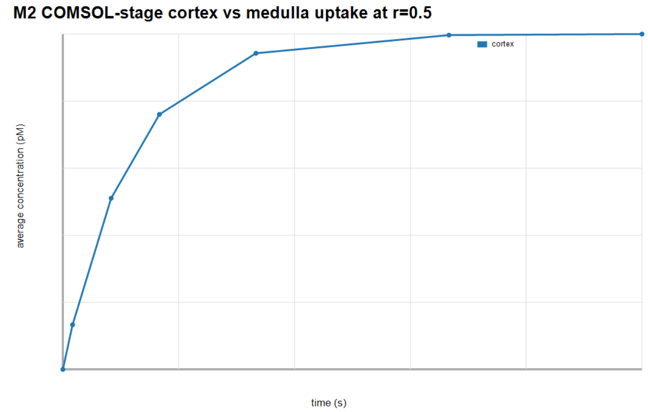

Fig. 7. Cortex, medulla, and sensor concentration trends at $r = 0.5$.

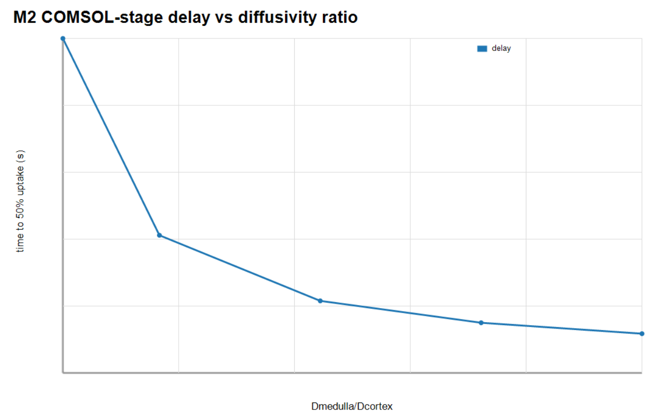

Fig. 8. Medulla uptake delay versus diffusivity ratio.

C. Experiment A: Flow versus diffusion

Experiment A compares the diffusion-only M2 baseline with M2B directional-flow cases at $v_{in} = 10^{-7}$, 5×10^{-7}, and 10^{-6} m/s for $C = 10$ pM. Directional flow improves sensor exposure: the reference 5×10^{-7} m/s case reaches 8.73 pM at 6000 s, compared with 4.98 pM for diffusion-only transport. The time to 50% sensor exposure decreases from approximately 6038 s under diffusion-only transport to approximately 1839 s for the reference flow case.

D. Experiment B: Sensor placement sensitivity

Sensor placement result. Experiment B evaluates three sensor placements: subcapsular/cortical, cortex-medulla transition, and medulla/hilum-side. Sensor placement affects response timing and current response. In the modeled sweep, the subcapsular/cortical placement is fastest and strongest under the reference flow case. Under directional flow, late-time exposure becomes high across all tested placements, while placement still changes arrival time and pathway exposure. The current-response range across placements is approximately 559.9–600.5 pA under flow and 348.2–533.3 pA under

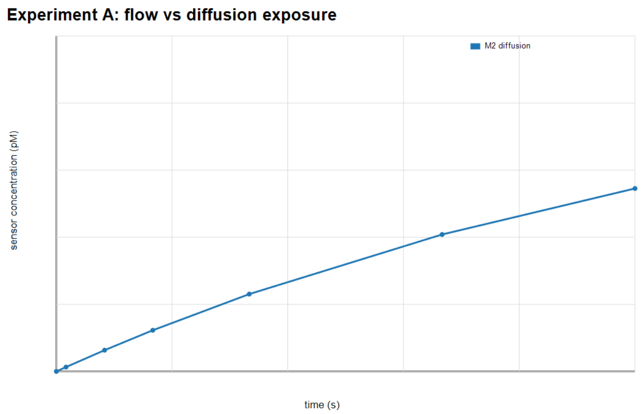

Fig. 9. Experiment A: sensor exposure under diffusion-only transport and directional-flow cases.

diffusion-only transport, showing that the late-time current spread is smaller under flow than diffusion-only transport.

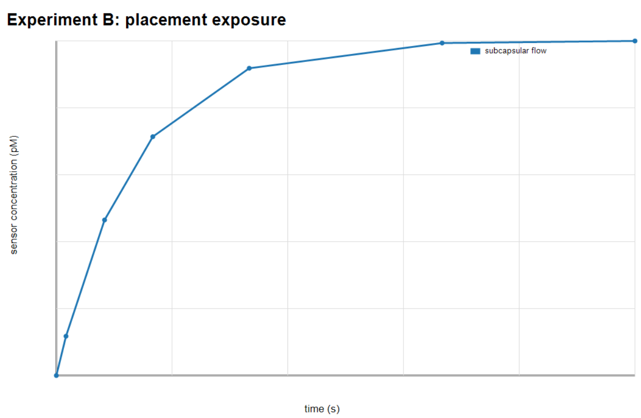

Fig. 10. Experiment B: concentration exposure for sensor placements under diffusion-only and directional-flow transport.

E. Experiment C: Detectability envelope

Electrical detectability result. Experiment C sweeps HER2 concentration, coupling efficiency, current noise floor, and binding affinity. Across the simulated envelope, 99 parameter combinations are detectable and 36 fail the 3σ current-threshold criterion. Detection is favored by stronger coupling, lower noise, lower K_d, and higher HER2 concentration. Detection fails for low concentration when coupling is weak, noise is high, or binding affinity is poor.

F. Experiment D: Debye screening feasibility

Debye screening feasibility result. Experiment D applies electrostatic attenuation using $\alpha_{eff} = \alpha_0 \exp(-d/\lambda_D)$. Screening strongly reduces feasibility as ionic strength increases and effective charge distance grows. For PBS-like physiological salt with $d \geq 5$ nm, none of the 60 screened

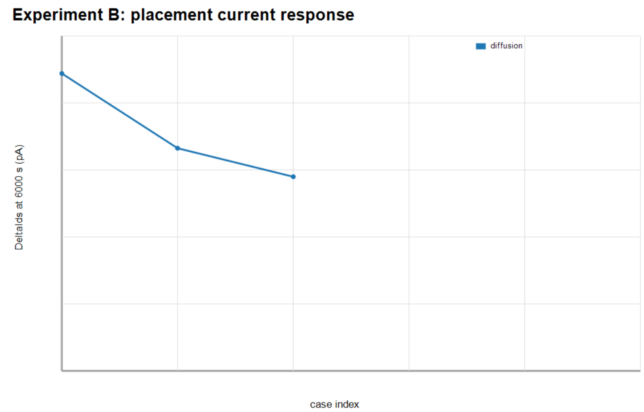

Fig. 11. Experiment B: GFET current response at 6000 s for the sensor-placement sweep.

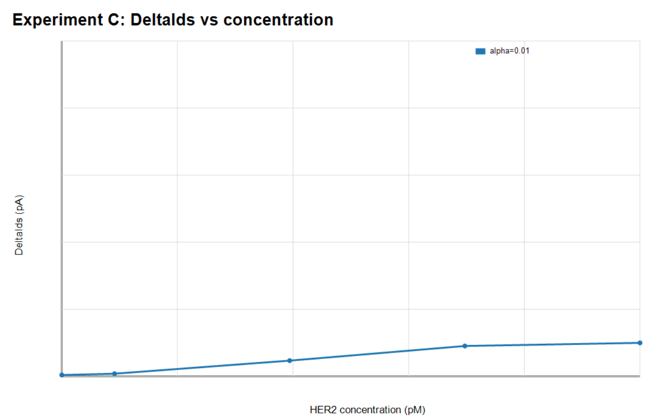

Fig. 12. Experiment C: current response versus HER2 concentration for coupling-efficiency cases.

cases remain detectable. This is a negative result: direct in-body GFET detection is not robust under PBS-like screening unless effective charge distance is very short, transduction is stronger, noise is lower, or the sensing environment reduces screening.

G. Surface Binding Response

Surface occupancy remains bounded and full-boundary exposure produces a larger total bound molecule count than the local GFET-scale sensing region.

H. DeltaIds and LOD

The current response increases with HER2 concentration and coupling efficiency. Under the current assumptions, the $\alpha = 0.03$, 10 pA case reaches a simulated LOD of approximately 0.83 pM. Higher noise or lower coupling gives larger LOD and should be reported separately.

V. Discussion

What was calculated: the project calculated HER2 binding occupancy, diffusion-only transport, partially anatomical

TABLE I
FINAL SCIENTIFIC CLAIMS SUPPORTED BY THE SIMULATION CAMPAIGN.

Claim	Supporting result	Figure/table	Assumption	Limitation	Confidence
HER2-antibody binding follows bounded saturation behavior across the modeled concentration range.	M1 occupancy increases monotonically with HER2 concentration and remains within [0, 1].	Fig. 1	Reversible Langmuir-type binding with baseline $K_d = 10$ pM.	No wet-lab binding calibration.	High for equation-level behavior.
Directional lymph-node-inspired flow changes sensor exposure compared with diffusion-only transport.	At $C = 10$ pM, $v_{in} = 5 \times 10^{-7}$ m/s gives 8.73 pM at 6000 s versus 4.98 pM for diffusion-only; time-to-50% exposure decreases from 6038 s to 1839 s.	Fig. 9	Prescribed velocity is coupled into TDS convection.	M2B is a partially anatomical prescribed-velocity convection-diffusion extension, not a full anatomical lymph-node model and not a validated Darcy-flow pressure solution.	Medium for model trend.
Sensor placement affects response timing and current response. In the modeled sweep, the subcapsular/cortical placement is fastest and strongest under the reference flow case. Under directional flow, late-time exposure becomes high across all tested placements, while placement still changes arrival time and pathway exposure.	Subcapsular/cortical placement is fastest and strongest under reference flow; late-time current range is 559.9–600.5 pA under flow and 348.2–533.3 pA under diffusion-only transport.	Fig. 10	Three placement regions are post-processed over the M2B transport model.	Placement sweep is not a full remeshed COMSOL sensor relocation study; late-time flow response is high across all tested placements.	Medium for relative sensitivity.
Detectability is predicted only under favorable coupling, low-noise, and binding assumptions.	EXP C gives 99 detectable rows and 36 failing rows across concentration, α_0, noise, and K_d sweeps.	Fig. 13	Analytical GFET response with $\Delta I_{DS} \geq 3\sigma$.	Graphene electronics and surface chemistry are simplified.	Medium for envelope; low for device claim.
PBS-like Debye screening strongly reduces effective GFET signal for long receptor distances.	For PBS-like $\lambda_D = 0.8$ nm and $d \geq 5$ nm, 0/60 screened cases remain detectable.	Fig. 14	$\alpha_{eff} = \alpha_0 \exp(-d/\lambda_D)$.	Debye lengths and distances are approximate assumptions.	Medium for screening direction.
Direct in-body HER2 detection likely requires short receptor distance, low-noise electronics, stronger transduction, or local screening mitigation.	Transport and placement improve antigen exposure, but detection fails under weak coupling, high noise, poor affinity, or PBS-like screening at longer distances.	Table I	Synthesis of M2B exposure, M3 binding, M4 response, and Debye attenuation.	Simulation-based design conclusion, not clinical validation.	High as modeling conclusion.

convection-diffusion transport, sensor-placement exposure, local bound molecule count, GFET current response, LOD, detectability envelopes, and Debye-screened current response.

What was concluded: antigen transport and placement influence how much HER2 reaches the sensor, but transport alone does not determine feasibility. Even when surface exposure improves, the predicted current can fall below the detection threshold if coupling is weak, noise is high, binding affinity is poor, or electrostatic screening is strong.

Positive result: directional lymph-node-inspired flow increases sensor exposure and reduces exposure time relative to diffusion-only transport under the modeled assumptions. Sensor placement affects response timing and current response. In the modeled sweep, the subcapsular/cortical placement is fastest and strongest under the reference flow case. Under directional flow, late-time exposure becomes high across all tested placements, while placement still changes arrival time and pathway exposure.

Negative result: detectability is not universal. Sub-pM or low-pM detection appears only under favorable coupling and low-noise assumptions, and PBS-like Debye screening with longer receptor distances eliminates detectability in the screened feasibility test.

What must improve for feasibility: the design would need shorter effective receptor-to-channel distance, lower-noise electronics, stronger transduction efficiency, higher-affinity recognition chemistry, or local mitigation of physiological ionic screening. These are engineering requirements suggested by the model, not demonstrated device performance.

VI. LIMITATIONS

This project is simulation-only. No wet-lab validation, fabrication, clinical testing, biocompatibility analysis, or direct physiological Debye screening model is included. The M2

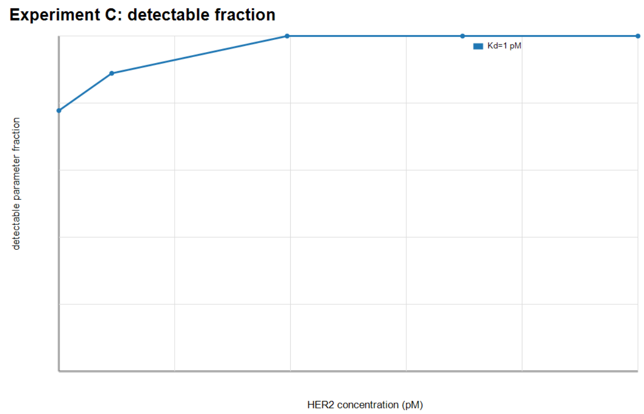

Fig. 13. Experiment C: detectable fraction across the concentration and affinity sweep.

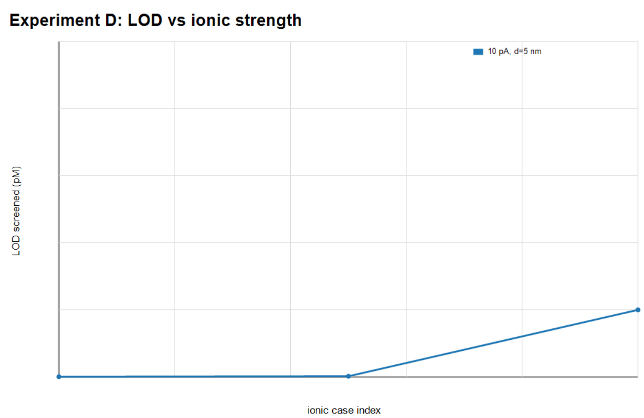

Fig. 14. Experiment D: screened LOD versus ionic-strength case.

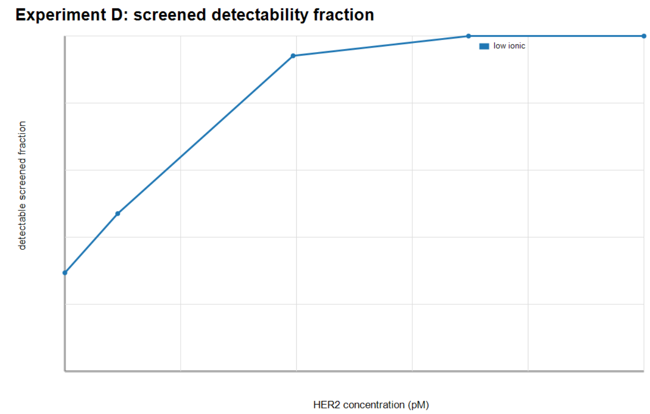

Fig. 15. Experiment D: screened detectability fraction under Debye attenuation assumptions.

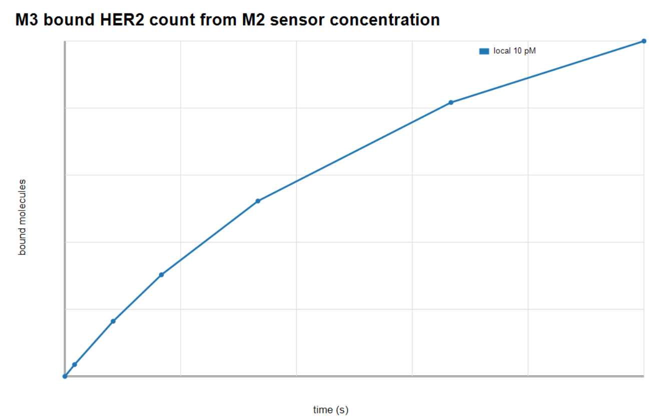

Fig. 16. Bound HER2 molecule count over time for local sensing at 10 pM.

model is meshed and solved in COMSOL, but direct field-table export from solved `.mph` files remains a repository limitation. M2B is a partially anatomical prescribed-velocity convection-diffusion extension, not a full anatomical lymph-node model and not a validated Darcy-flow pressure solution. Direct COMSOL field-table export remains future work; the current quantitative tables are post-processed from the documented COMSOL-stage model and parameter sweeps. Surface binding and GFET response are analytically post-processed rather than fully coupled inside COMSOL.

VII. Conclusion

The main scientific conclusion of this project is that HER2 transport alone does not determine GFET biosensor feasibility. Even when antigen reaches the sensor, electrical detectability depends strongly on coupling efficiency, noise floor, and electrostatic screening.

The repository now implements a staged HER2-GFET workflow: 0D binding, meshed COMSOL transport, a separate partially anatomical prescribed-velocity convection-diffusion extension, surface binding from sensor concentration, and GFET current-response/LOD estimation. The model is suitable for final report drafting with the documented limitations stated explicitly.

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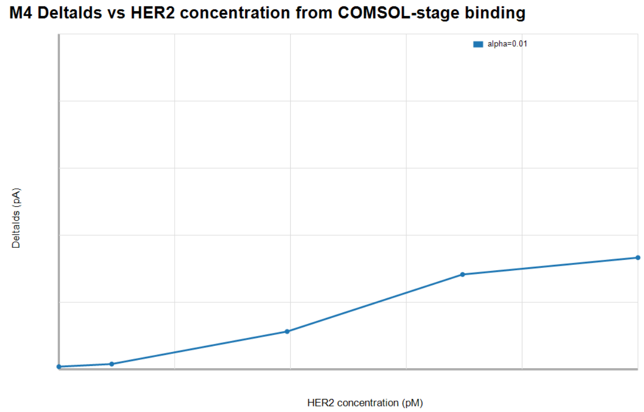

Fig. 17. GFET current response versus HER2 concentration.

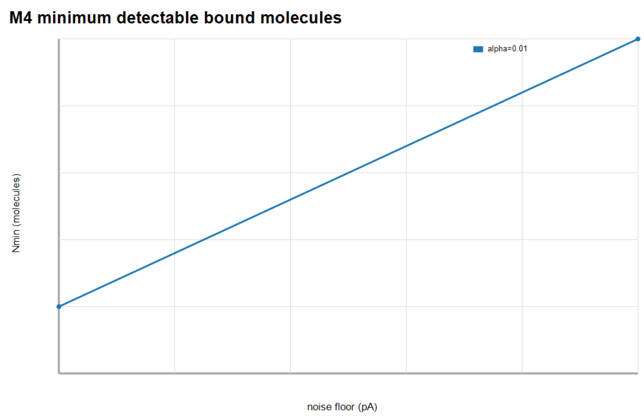

Fig. 18. Minimum detectable bound molecule count for coupling and noise-floor cases.

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